Open Source Software Audit Report

Software: Git

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Course: Open Source Software

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Git was created by Linus Torvalds in 2005. Before Git, developers working on the Linux kernel were using a proprietary version control system called BitKeeper. However, due to licensing issues, they lost access to it.

This created a major problem because the Linux kernel project involved thousands of developers working together, and they needed a fast and reliable system to manage code changes.

Linus Torvalds decided to build a new version control system that would be free and open source. His goal was to create a tool that was fast, efficient, and capable of handling large projects.

Git was designed with a distributed model, meaning every developer has a complete copy of the project. This makes it more flexible and powerful compared to older systems.

Today, Git is one of the most widely used version control systems and is essential for modern software development.

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```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\IPSITA> wsl --install
The requested operation requires elevation.
Downloading: Windows Subsystem for Linux 2.6.3
Installing: Windows Subsystem for Linux 2.6.3
Windows Subsystem for Linux 2.6.3 has been installed.
The operation completed successfully.
Downloading: Ubuntu
Installing: Ubuntu
Distribution successfully installed. It can be launched via 'wsl.exe -d Ubuntu'
Launching Ubuntu...
Provisioning the new WSL instance Ubuntu
This might take a while...
Create a default Unix user account: ipsita
New password:
Retype new password:
passwd: password updated successfully
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ipsita@LAPTOP-P36DNI7V:/mnt/c/Users/IPSITA$ sudo apt update
[sudo] password for ipsita:
Sorry, try again.
[sudo] password for ipsita:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1551 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [249 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.7 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1166 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [225 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.9 kB]
```

Search

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Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.9 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2701 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [627 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [544 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [28.8 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6980 B]
Get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:21 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Get:22 http://archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1847 kB]
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [339 kB]
Get:31 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:32 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:33 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1666 kB]
Get:34 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [323 kB]
Get:35 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:36 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.2 kB]
Get:37 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2837 kB]
Get:38 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [656 kB]
Get:39 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:40 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [556 B]
Get:41 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.1 kB]
Get:42 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7520 B]
Get:43 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:44 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [500 B]
Get:45 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]
Get:46 http://archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9208 B]
Get:47 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7368 B]
Get:48 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:49 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.7 kB]
Get:50 http://archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.2 kB]
Get:51 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.2 kB]
Get:52 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1480 B]
Get:53 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
```

Search

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Get:53 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:54 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
Get:55 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [780 B]
Get:56 http://archive.ubuntu.com/ubuntu noble-backports/multiverse Translation-en [372 B]
Get:57 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:58 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 c-n-f Metadata [116 B]
Fetched 41.1 MB in 40s (1033 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
63 packages can be upgraded. Run 'apt list --upgradable' to see them.
ipsita@LAPTOP-P36DNI7V:/mnt/c/Users/IPSITA$ git --version
git version 2.43.0
ipsita@LAPTOP-P36DNI7V:/mnt/c/Users/IPSITA$ git --version
git version 2.43.0
ipsita@LAPTOP-P36DNI7V:/mnt/c/Users/IPSITA$ which git
/usr/bin/git
ipsita@LAPTOP-P36DNI7V:/mnt/c/Users/IPSITA$ git config --global user.name "Ipsita Singh"
git config --global user.email "ipsita@email.com"
git config --list
user.name=Ipsita Singh
user.email=ipsita@email.com
ipsita@LAPTOP-P36DNI7V:/mnt/c/Users/IPSITA$ mkdir testrepo
cd testrepo
git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
error: chmod on /mnt/c/Users/IPSITA/testrepo/.git/config.lock failed: Operation not permitted
fatal: could not set 'core.filemode' to 'false'
ipsita@LAPTOP-P36DNI7V:/mnt/c/Users/IPSITA/testrepo$ cd ~
ipsita@LAPTOP-P36DNI7V:~$ ipsita@LAPTOP:~$
ipsita@LAPTOP:~$: command not found
ipsita@LAPTOP-P36DNI7V:~$ mkdir testrepo
```

ipsita@LAPTOP-P36DNI7V: ~/

```
user.email=ipsita@email.com
ipsita@LAPTOP-P36DNI7V:/mnt/c/Users/IPSITA$ mkdir testrepo
cd testrepo
git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
error: chmod on /mnt/c/Users/IPSITA/testrepo/.git/config.lock failed: Operation not permitted
fatal: could not set 'core.filemode' to 'false'
ipsita@LAPTOP-P36DNI7V:/mnt/c/Users/IPSITA/testrepo$ cd ~
ipsita@LAPTOP-P36DNI7V:~$ ipsita@LAPTOP:~$
ipsita@LAPTOP:~$: command not found
ipsita@LAPTOP-P36DNI7V:~$ mkdir testrepo
cd testrepo
git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/ipsita/testrepo/.git/
ipsita@LAPTOP-P36DNI7V:~/testrepo$ git status
ls -a
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
.  ..  .git
```